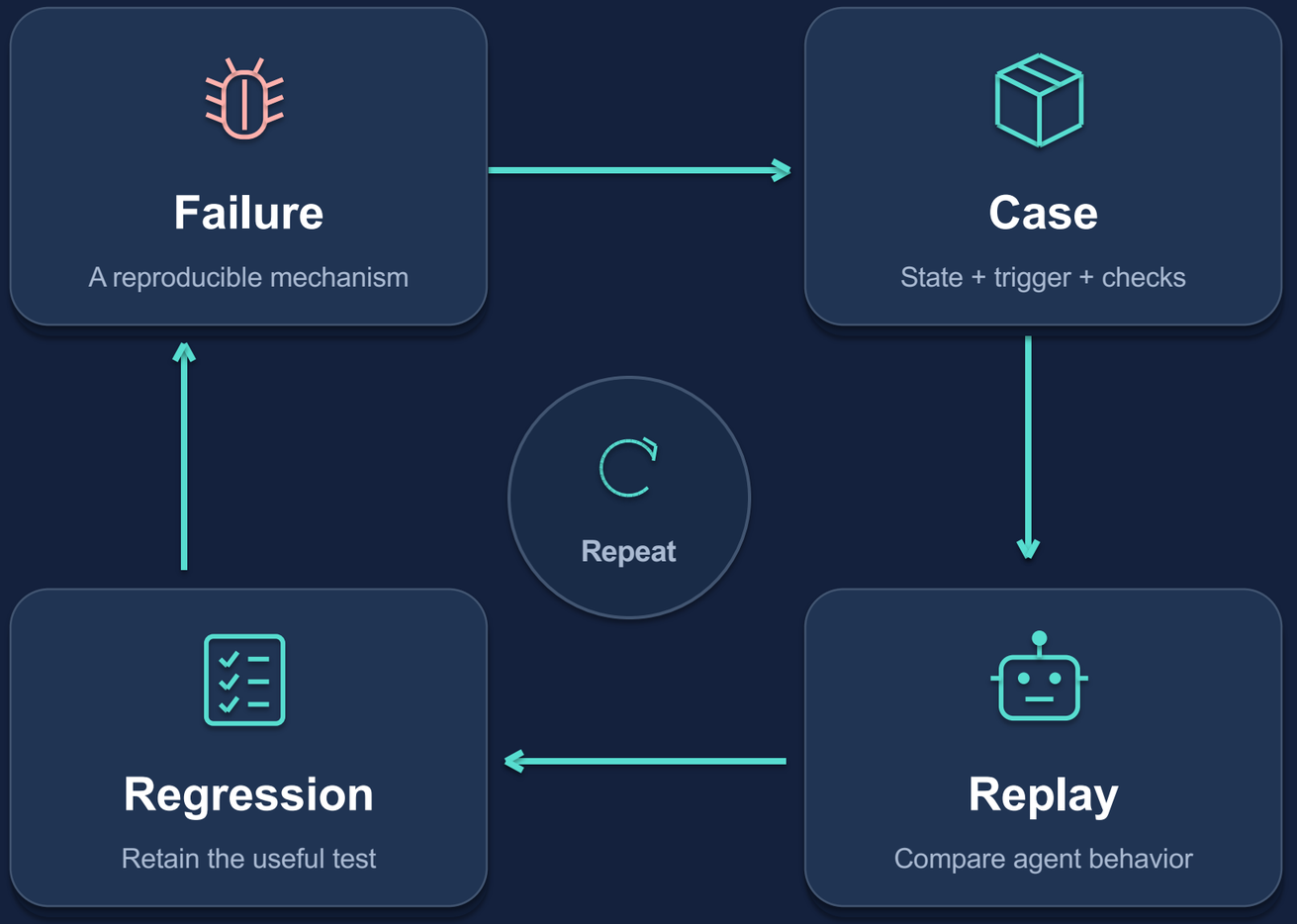
Incident Replay

From software failures to repeatable engineering tests.

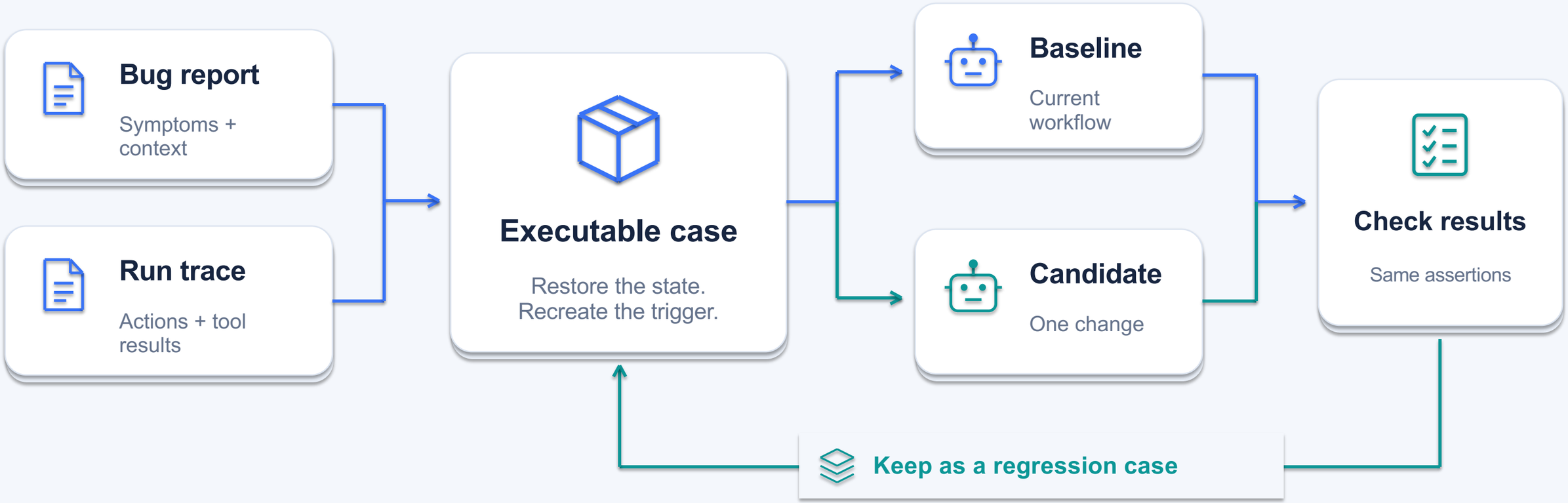
SWE AGENT EVALUATION

Jiawei Li



Turn a failure into a repeatable experiment.

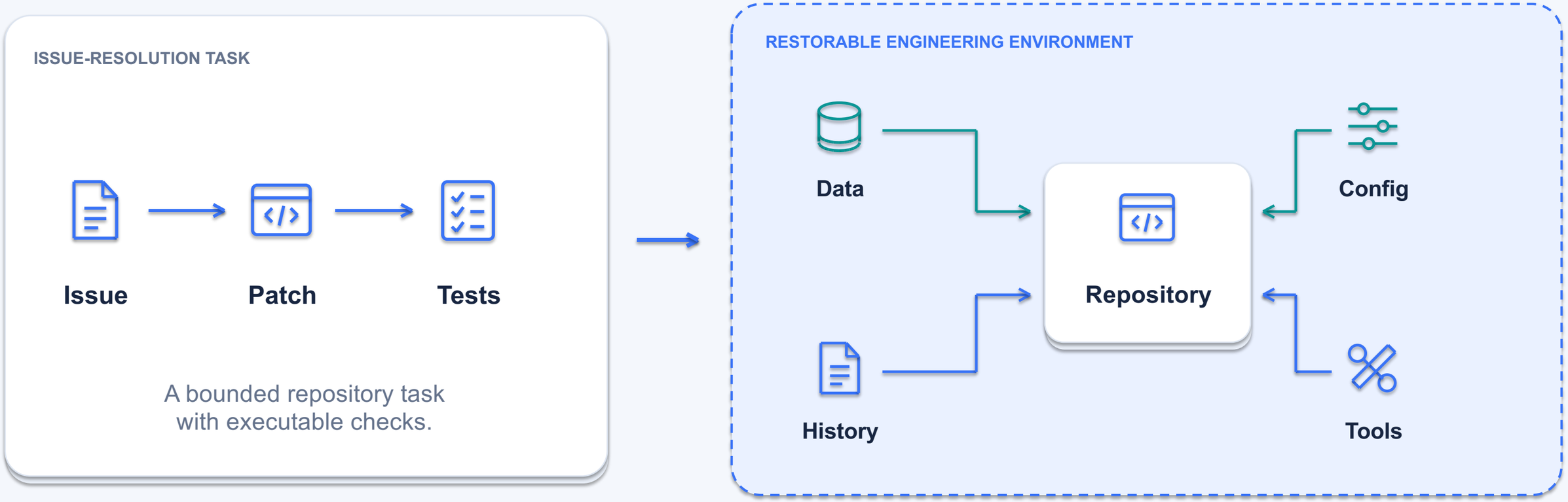
Capture the conditions, compare a change, and keep the case for the next release.



A report becomes an executable test.

Expand the test boundary around the patch.

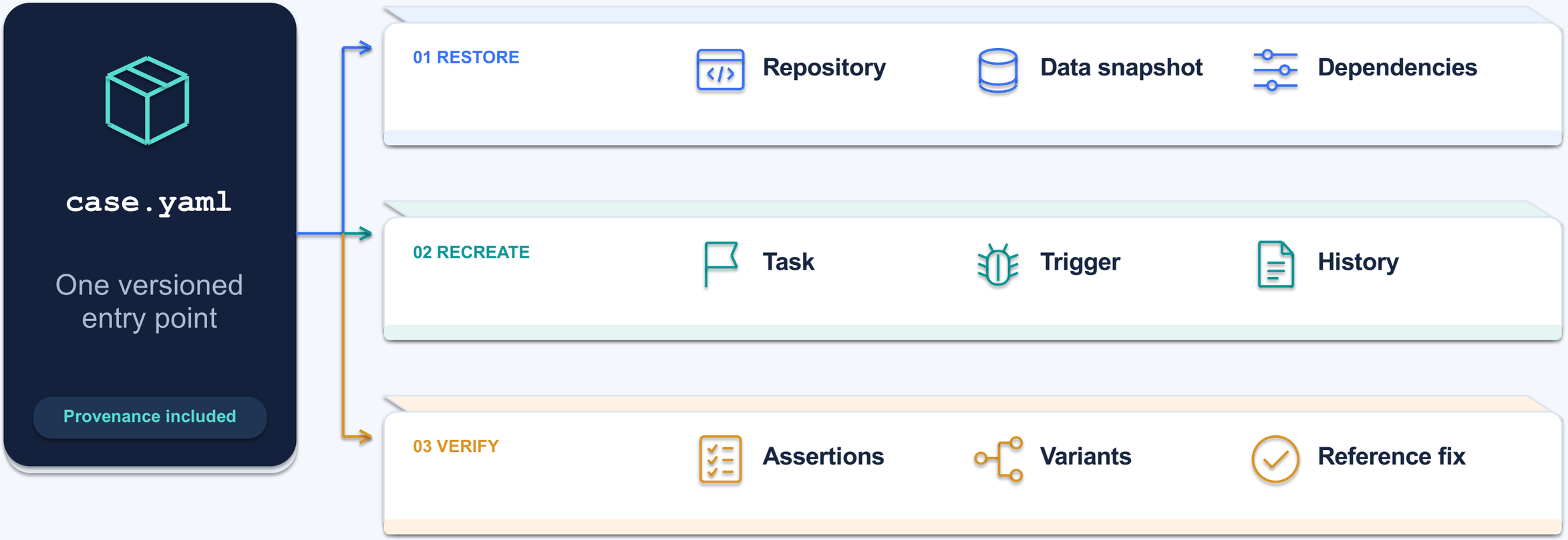
Incident replay makes operational state and recovery explicit.



SWE-bench already uses real repository issues. Replay adds explicit state, action history, and recovery criteria.

A case packages the complete experiment.

Three connected layers make the failure inspectable, resettable, and reusable.



Three replay modes. Three different questions.

The execution path determines what the comparison can establish.

01

Inspect a trace

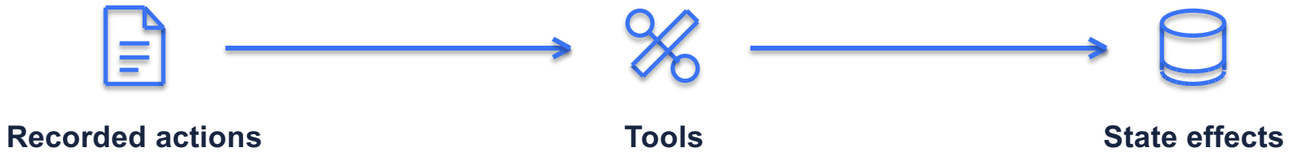
What happened?



02

Replay actions

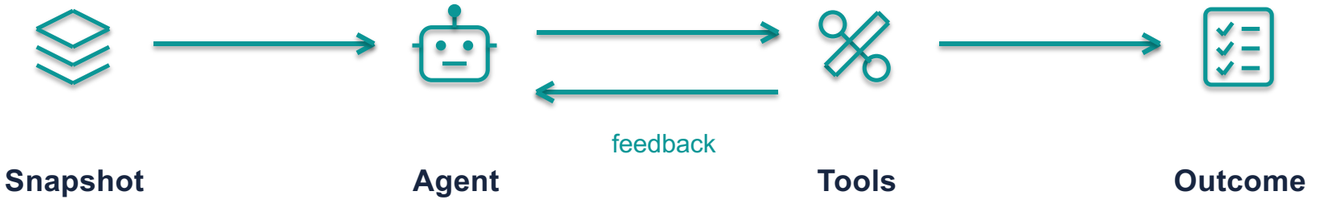
What changes the state?



03

Re-run the agent

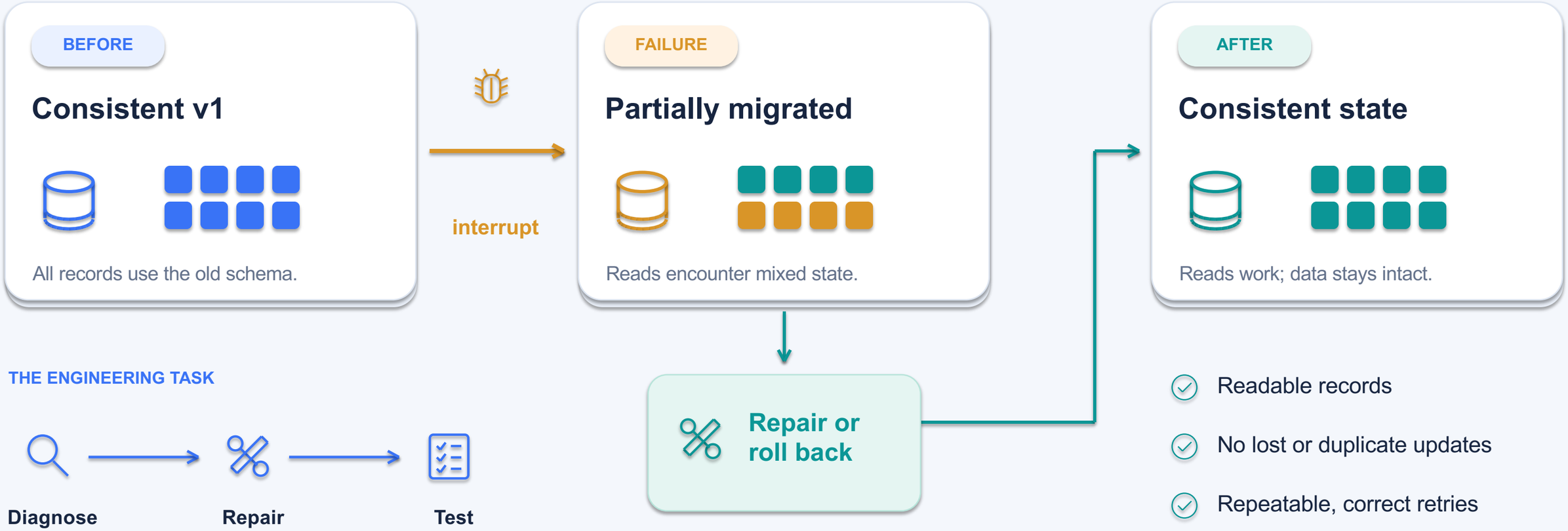
Can it solve the task now?



Changed observations require new agent decisions; fixed actions cannot show adaptation.

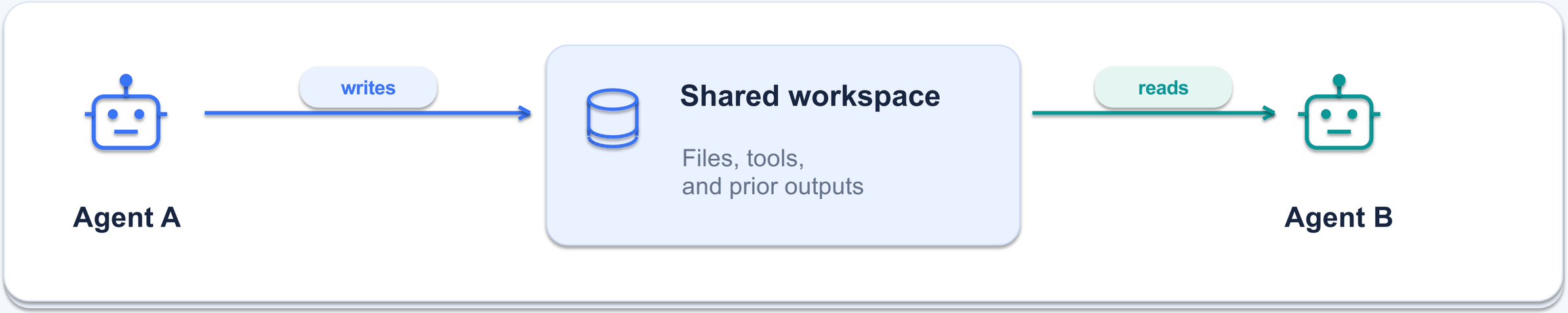
Replay the state transition behind a failure.

An interrupted migration becomes a task in diagnosis, data repair, and regression testing.



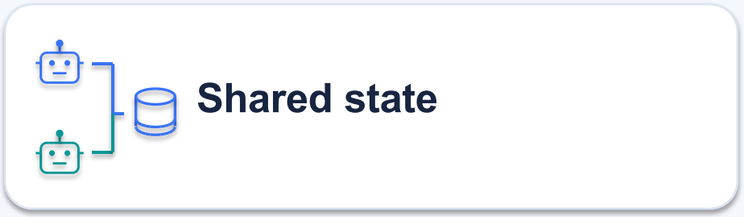
One agent’s output becomes another’s input.

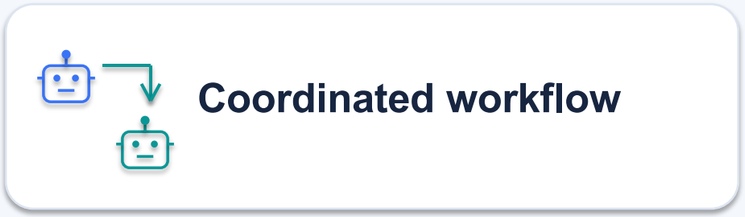
Shared infrastructure creates dependencies between otherwise separate trajectories.



PROPOSED COMPARISONS: CHANGE THE SHARING TOPOLOGY



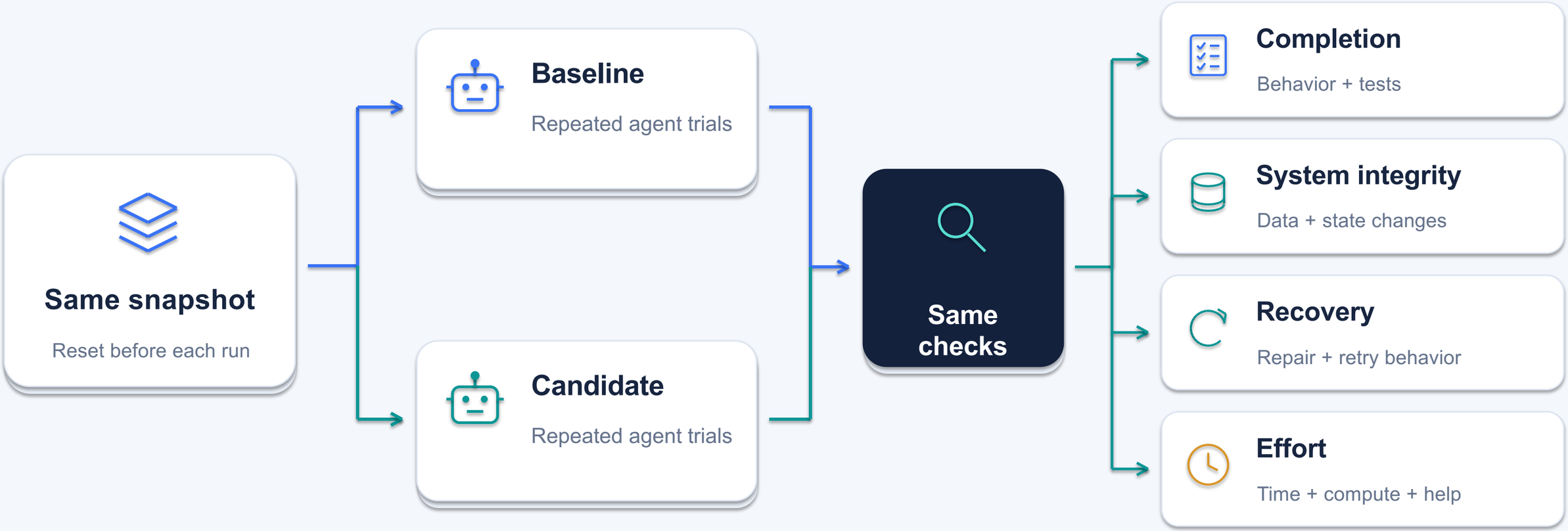




Published component environments are the starting point; a connected replay is further work.

Compare the change from the same starting state.

Keep four outcome dimensions visible, alongside repeated trials and matched budgets.



Also test normal cases; report uncertainty and unsuccessful runs.

Grow coverage by failure mechanism.

Start with one qualified family, then add different sources of engineering failure.

01



Shared state

OpenAI–HF component environments

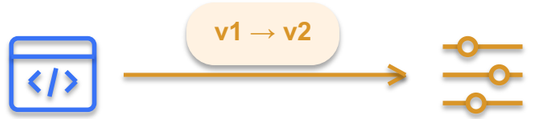
02



Partial updates

Migration or failed recovery

03



Environment drift

Dependency or configuration mismatch

BUILD ORDER: MOVE FORWARD WHEN THE CASE IS READY

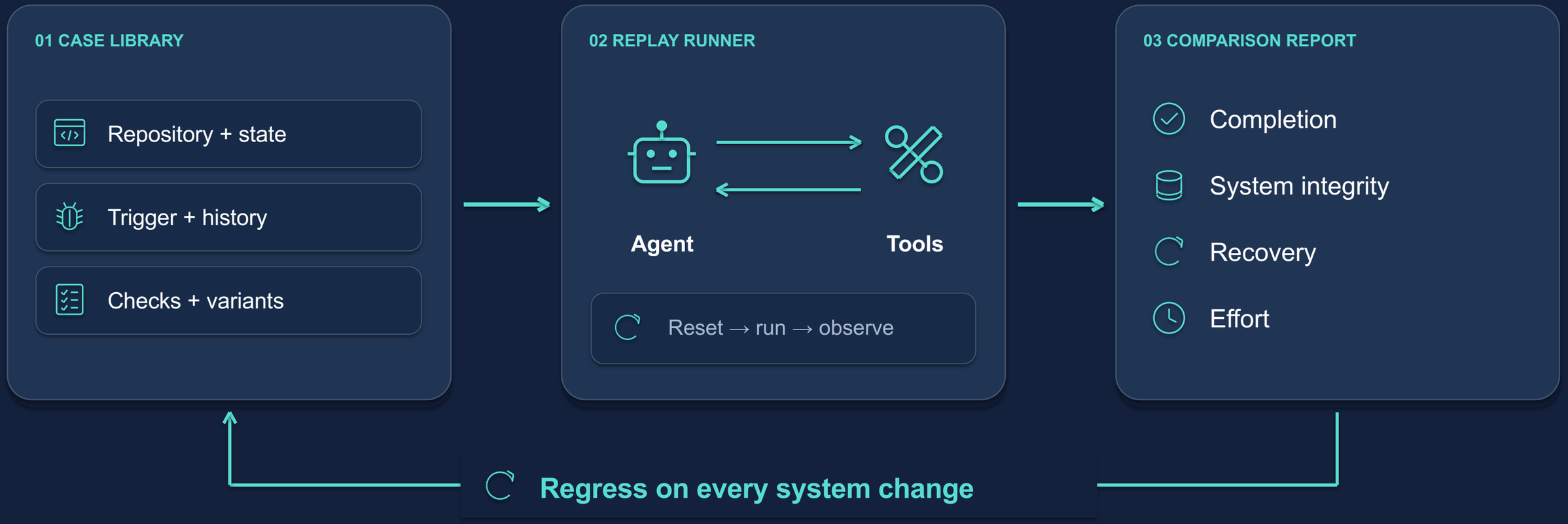
Qualify fixture

Compare changes

Retain as regression

A regression loop that grows with every failure.

Ship the case library, replay runner, and evidence together.



Every reproducible failure becomes reusable engineering evidence.